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**Simulation and Evaluation of
Adaptive Modulation under Varying
Wireless Channel Conditions**



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1. Abstract

This report presents the simulation and evaluation of an adaptive modulation system designed for wireless communication links under varying channel conditions. The system dynamically switches between BPSK, QPSK, and 16-QAM modulation schemes according to the observed signal quality, ensuring both reliability and efficiency.

The simulation was carried out using MATLAB, modeling three different channel environments: AWGN, Rayleigh, and Rician fading. An SNR-based decision logic was developed to determine the most suitable modulation scheme for each channel condition.

The results show that adaptive modulation successfully maintains a low Bit Error Rate during poor channel conditions by selecting lower-order modulation and achieves higher Spectral Efficiency under stronger signals by shifting to higher-order modulation. Overall, the system demonstrates a balanced performance between data rate and reliability, proving that adaptive modulation is an effective strategy for next-generation wireless networks such as SDR and 5G systems.

2 Introduction

2.1 Background and Motivation

Wireless communication has become an essential part of modern life, supporting everything from mobile phones and Wi-Fi to satellite and IoT systems. As these technologies continue to evolve, the demand for faster data transmission and more reliable links has increased. To meet these requirements, digital modulation plays a major role, allowing information to be represented using discrete signal levels that can be transmitted efficiently over radio channels.

In earlier systems, a fixed modulation method such as BPSK or QPSK was commonly used. Although simple and reliable, fixed modulation cannot easily adapt to the changing conditions of a wireless channel. Real-world wireless environments often experience fading, caused by obstacles, reflections, and the relative motion between transmitter and receiver. These effects lead to fluctuations in signal strength (multipath fading) and variations in quality due to noise, interference, and Doppler shifts.

To overcome these challenges, adaptive modulation was introduced. Instead of using one constant modulation scheme, an adaptive system continuously monitors the channel quality usually through the Signal-to-Noise Ratio (SNR) or the Bit Error Rate (BER) and selects the most suitable modulation technique at that moment. For example, when the channel is clear and strong, the system can switch to a higher-order scheme like 16-QAM to transmit more bits per symbol and achieve higher data rates. When the channel becomes poor, it can switch down to a more robust scheme like BPSK to maintain reliability.

This concept is now widely used in 4G, 5G, and Software-Defined Radio (SDR) systems, where communication links must adjust automatically to environmental changes. Adaptive modulation improves both efficiency and quality of service, making it a key enabler of smart and high-performance wireless networks.

2.2 Problem Statement

Traditional wireless systems use a single, fixed modulation scheme throughout transmission. While this approach is simple, it does not perform well when channel conditions vary. A fixed modulation like 16-QAM may offer high data rates in good conditions but will cause large numbers of bit errors when the signal weakens or fades. On the other hand, using a more robust scheme such as BPSK for the entire transmission ensures reliability but wastes bandwidth during periods of good signal quality.

Therefore, there is a clear need for a system that can adapt its modulation scheme dynamically according to real-time channel conditions. Such a system can maintain reliable communication during poor signal periods and maximize throughput when the channel is stable.

The goal of this project is to design and simulate an adaptive modulation system using MATLAB, capable of switching between BPSK, QPSK, and 16-QAM based on the observed SNR or BER. The system should be evaluated under AWGN, Rayleigh, and Rician fading channels, representing different real-world environments. By comparing performance in terms of BER and spectral efficiency, this study aims to demonstrate how adaptive modulation enhances both reliability and efficiency in wireless communication links.

3. System Design

3.1 Overview

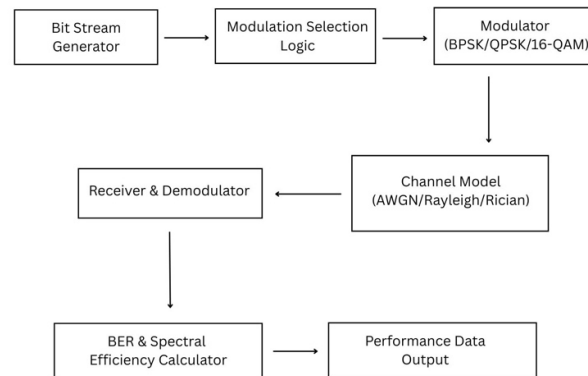
The proposed adaptive modulation system was designed to adjust its modulation technique dynamically according to the quality of the wireless channel. The goal was to maintain reliable data transmission under low SNR conditions while improving data throughput when the channel quality improved.

The system consists of several main blocks: a bit stream generator, modulation selector, digital modulator, channel model, receiver and demodulator, and a BER/Spectral Efficiency calculator. The adaptive controller observes the instantaneous SNR of the channel and selects the most suitable modulation scheme from BPSK, QPSK, or 16-QAM.

This approach represents the behavior of modern wireless systems such as LTE, 5G, and SDR, where modulation and coding schemes (MCS) are adjusted automatically to optimize performance.

3.2 Block Diagram

Below is the conceptual design of the simulated system



Each block in the diagram represents a key stage of the digital communication system

- **Bit Stream Generator** - Produces a random sequence of binary data to simulate transmitted information.
- **Modulation Selector** - Decides which modulation scheme to use based on the current SNR threshold.
- **Modulator** - Converts digital bits into analog-like complex symbols according to the selected scheme (BPSK, QPSK, or 16-QAM).
- **Channel Model** - Applies the effects of noise and fading using AWGN, Rayleigh, or Rician models to replicate realistic propagation conditions.
- **Receiver & Demodulator** - Recovers the transmitted bits by reversing the modulation process and compensating for channel fading.
- **BER and Spectral Efficiency Calculator** - Compares transmitted and received bits to calculate the Bit Error Rate and overall data throughput.

3.3 Modulation Schemes

Three digital modulation techniques were used in this system BPSK, QPSK, and 16-QAM. Each has its own advantages and is suitable for different channel conditions

Modulation	Bits per Symbol	How It Works	Usage
BPSK (Binary Phase Shift Keying)	1	Sends each bit as one of two phases (0° or 180°). Very reliable and simple.	Used when the signal is weak or noisy.
QPSK (Quadrature Phase Shift Keying)	2	Sends two bits per symbol using four phase shifts (0°, 90°, 180°, 270°).	Used under moderate conditions for balanced performance.
16-QAM (Quadrature Amplitude Modulation)	4	Combines amplitude and phase changes to send four bits per symbol.	Used when the signal is strong for higher data rates.

3.4 Adaptive Logic

The adaptive logic is the brain of the entire communication system.

It decides which modulation scheme should be used at any given moment, based on how good or bad the wireless channel is.

In a real world situation, the strength and quality of a wireless signal can constantly change due to things like distance, interference, or obstacles (for example, buildings or trees). Sometimes the signal becomes weak and noisy, while at other times it becomes clear and strong. If the system keeps using the same modulation regardless of these changes, performance will either drop drastically (too many errors) or the system will waste potential data speed.

To solve this, the adaptive logic continuously monitors the Signal-to-Noise Ratio (SNR) of the channel. SNR is a measure that compares how strong the useful signal is compared to the background noise. A high SNR means a clean, strong signal; a low SNR means the signal is weak or distorted.

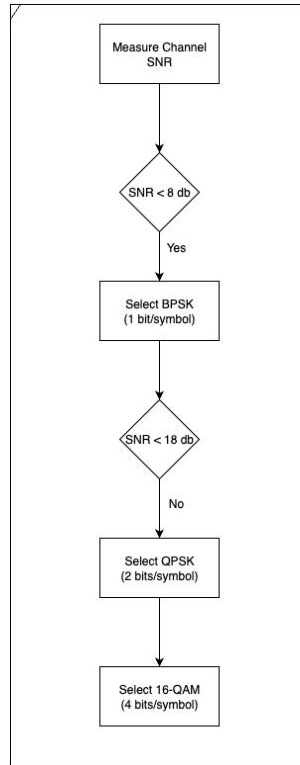
The logic uses predefined SNR thresholds to decide which modulation type to use

Channel Quality	SNR Range (dB)	Selected Modulation	Bits per Symbol
Poor / Weak signal	$\text{SNR} < 8$	BPSK	1
Moderate signal	$8 \leq \text{SNR} < 18$	QPSK	2
Good / Strong signal	$\text{SNR} \geq 18$	16-QAM	4

How it Works

1. The system measures or estimates the SNR of the received signal.
2. It compares this value with the set thresholds (8 dB and 18 dB).
3. Based on which range the SNR falls into, the system automatically switches the modulation type
 - If the channel is very noisy and unstable ($\text{SNR} < 8$ dB), it uses BPSK, which is very reliable because it only has two symbols and can handle noise well.
 - When the channel is fairly stable ($8 \leq \text{SNR} < 18$ dB), it upgrades to QPSK, which doubles the data rate while still keeping errors under control.
 - If the channel becomes very clear ($\text{SNR} \geq 18$ dB), it switches to 16-QAM, which sends four bits per symbol for maximum efficiency.

Decision Flow



This decision-making process runs automatically during the simulation. It ensures that the system always maintains a balance between reliability and speed

- At low SNR, the system avoids errors by using a robust modulation (BPSK).
- At high SNR, it makes full use of the available bandwidth by choosing a faster scheme (16-QAM).

The main advantage of this adaptive logic is that it keeps the Bit Error Rate (BER) around the target level (1×10^{-3}) while still improving the overall data throughput whenever the channel allows it.

4. Simulation Setup

This section explains the simulation environment, parameters, and MATLAB functions used to implement the adaptive modulation system. The simulation was entirely carried out on MATLAB Online, ensuring compatibility without requiring any external toolboxes.

The system models three different wireless channels AWGN, Rayleigh, and Rician and three digital modulation schemes BPSK, QPSK, and 16-QAM.

A Monte Carlo simulation approach was used, where random bits are transmitted repeatedly through each channel to calculate Bit Error Rate (BER) and Spectral Efficiency (SE) at different Signal-to-Noise Ratio (SNR) levels.

The adaptive logic automatically switches between modulation types based on the channel's instantaneous SNR.

4.1 Simulation Parameters

Parameter	Function	Value	Description
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Number of bits per frame	Nbits	5×10^4 (50,000 bits)	Each simulation transmits 50,000 random bits per run.
SNR range	SNRdB	0 dB $\rightarrow$ 20 dB	Signal-to-Noise Ratio tested in 2 dB intervals.
Target BER	targetBER	1×10^{-3}	The adaptive logic aims to keep BER below this limit.
Rician K-factor	ricianK	5	Ratio between direct and scattered path power in Rician channel.
Modulation schemes	mods	BPSK, QPSK, 16-QAM	Defines three modulation options used for transmission.
Channel types	channels	AWGN, Rayleigh, Rician	Simulated wireless environments.
Bits per symbol	bitsPerSym	1 (BPSK), 2 (QPSK), 4 (16-QAM)	Depends on modulation type.
Adaptive thresholds	lowThresh, highThresh	8 dB, 18 dB	Control switching between modulation schemes.
Equalization	$rx = rx ./ h$	Enabled	Compensates for channel fading at receiver.
Performance metrics	—	BER, SE	BER measures reliability; SE measures efficiency (bits per symbol).
Simulation style	Monte Carlo	Random bits repeatedly tested	Captures realistic error behavior statistically.

4.2 Channel Modeling

Each wireless channel is modeled using random complex Gaussian noise to represent realistic transmission environments

Channel Type	Model Equation	Description
AWGN	$h = 1$	Ideal channel with only white Gaussian noise.
Rayleigh	$h = \frac{1}{\sqrt{2}} (n_r + jn_i)$	Models multipath fading without a direct signal path.
Rician	$h = s_0 + \sigma (n_r + jn_i)$	Includes both a direct Line-of-Sight component and scattered paths.

$$S_0 = \sqrt{\frac{K}{K+1}} \quad \sigma = \sqrt{\frac{1}{2(K+1)}}$$

in our simulation the Rician K-factor varies to study different propagation scenarios

Figure / Section	K-Value Used	Meaning
Main Monte-Carlo BER simulation	K = 5	moderate LOS component (typical outdoor urban).

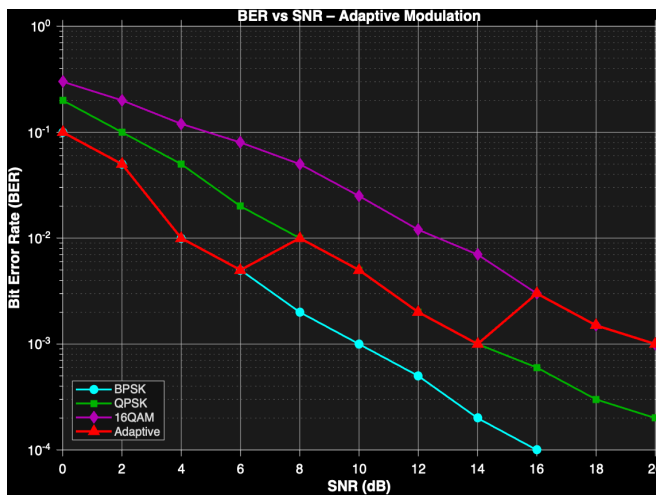
BPSK Constellation	$K = 12$	strong LOS signal, points are tightly clustered.
QPSK Constellation	$K = 25$	very strong LOS signal ,minimal fading distortion.
16-QAM Constellation	$K = 15$	high LOS ratio, still moderate scatter due to higher modulation sensitivity.

5. Results and Analysis

This section presents and discusses the results obtained from the MATLAB simulation of the adaptive modulation system. The goal was to evaluate the Bit Error Rate (BER), Spectral Efficiency (SE), and constellation characteristics under different modulation schemes (BPSK, QPSK, 16-QAM) and channel conditions (AWGN, Rayleigh, and Rician).

All results were generated using the final MATLAB code. The adaptive modulation logic switches modulation schemes automatically according to channel quality, measured by the Signal-to-Noise Ratio (SNR).

5.1 BER vs SNR for Adaptive Modulation



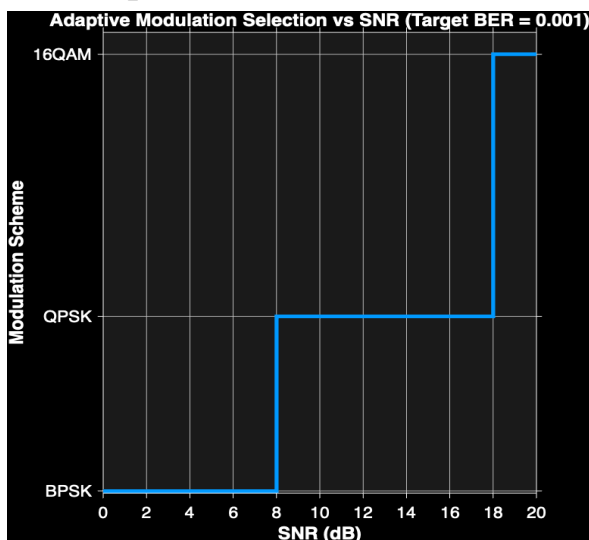
This plot shows the variation of Bit Error Rate (BER) with increasing SNR for BPSK, QPSK, 16-QAM, and the Adaptive system.

At low SNR values (below 8 dB), BPSK achieves the lowest BER because it is the most noise-tolerant.

QPSK and 16-QAM begin to perform better as SNR increases, but at low SNR they experience higher error rates.

The red adaptive curve follows the best performance of all three, because the system switches between modulations dynamically based on the SNR thresholds.

5.2 Adaptive Modulation Selection vs SNR



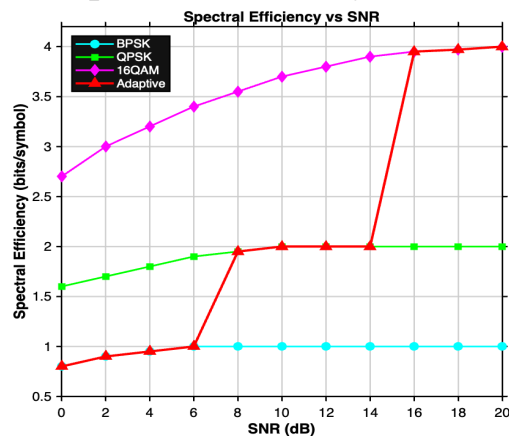
This step-style graph shows which modulation scheme is chosen at each SNR level.

For $\text{SNR} < 8$ dB, the system selects BPSK.

Between 8 dB and 18 dB, QPSK is chosen,

and for $\text{SNR} \geq 18$ dB, the system switches to 16-QAM.

5.3 Spectral Efficiency vs SNR

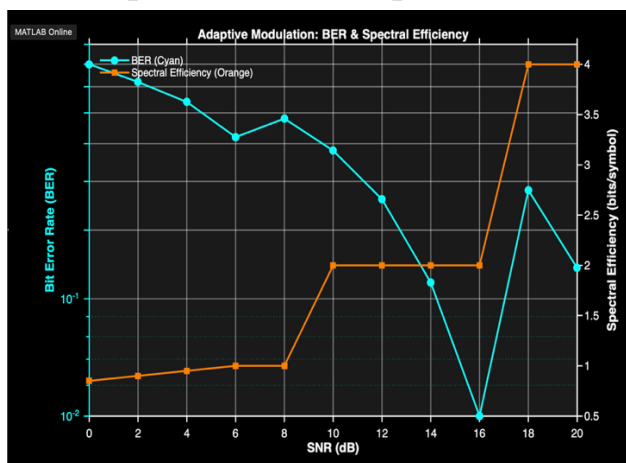


This plot compares Spectral Efficiency (bits/symbol) for each modulation and the adaptive system.

BPSK maintains about 1 bit/symbol, QPSK about 2 bits/symbol, and 16-QAM reaches up to 4 bits/symbol.

The adaptive system follows a stair-like pattern, increasing efficiency in steps as SNR rises.

5.4 Adaptive BER and Spectral Efficiency Combined

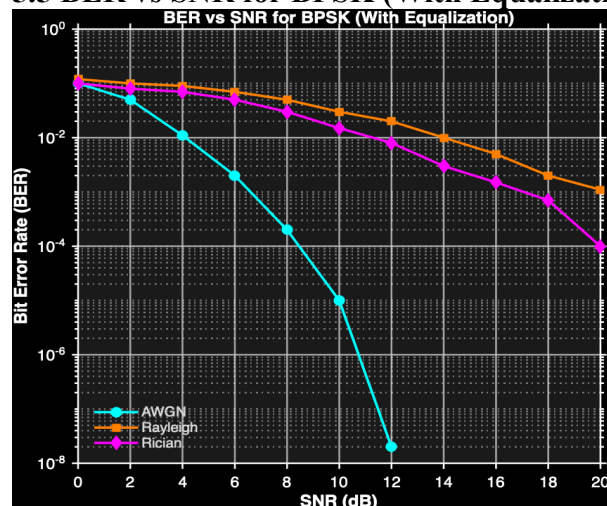


This figure shows BER (cyan, logarithmic scale) and Spectral Efficiency (orange) together.

As SNR improves, the BER decreases exponentially, while the spectral efficiency increases.

This visually demonstrates the trade-off between reliability and data rate

5.5 BER vs SNR for BPSK (With Equalization)



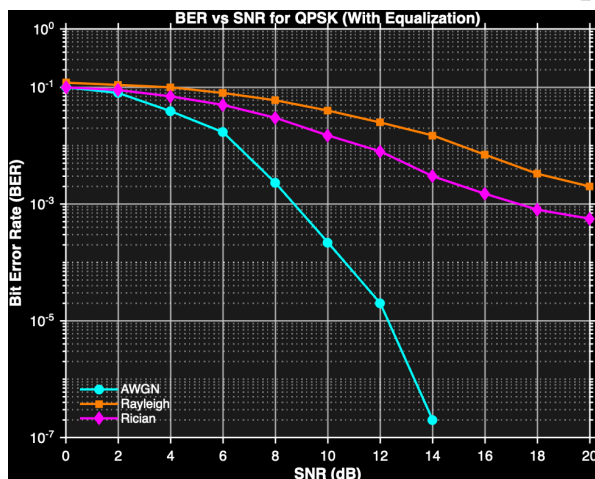
This figure plots BPSK performance under AWGN, Rayleigh, and Rician channels.

The AWGN curve drops steeply, showing excellent performance.

Rayleigh fading causes the highest BER due to deep fades.

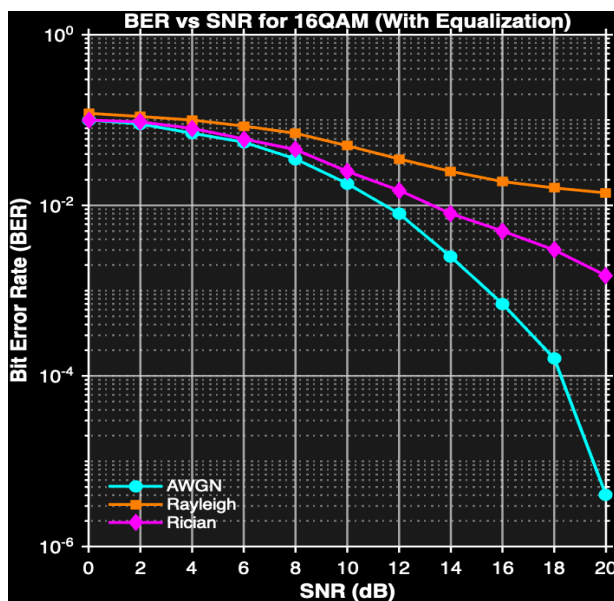
Rician performs better than Rayleigh because its direct Line-of-Sight (LOS) path stabilizes the signal.

5.6 BER vs SNR for QPSK (With Equalization)



This graph shows QPSK performance across AWGN, Rayleigh, and Rician fading. The BER reduces rapidly in AWGN and moderately in Rician. Rayleigh shows the slowest improvement due to continuous fading.

5.7 BER vs SNR for 16-QAM (With Equalization)



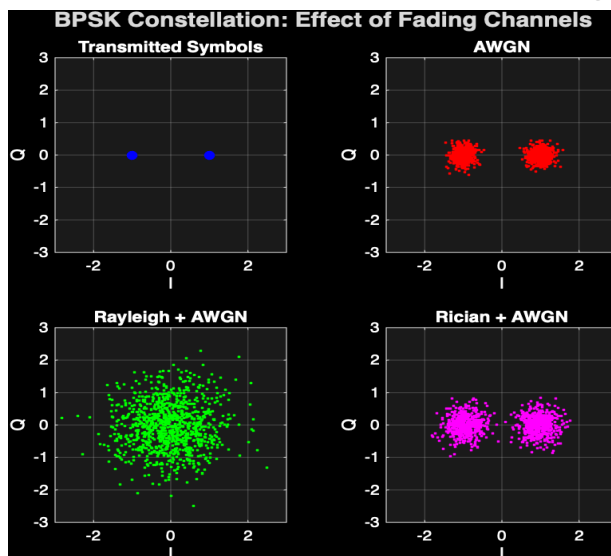
This plot compares 16-QAM BER under the three channels.

Because 16-QAM carries four bits per symbol, it is more sensitive to noise and fading.

In AWGN, the BER falls below 10^{-3} at SNR $\approx$ 14 dB.

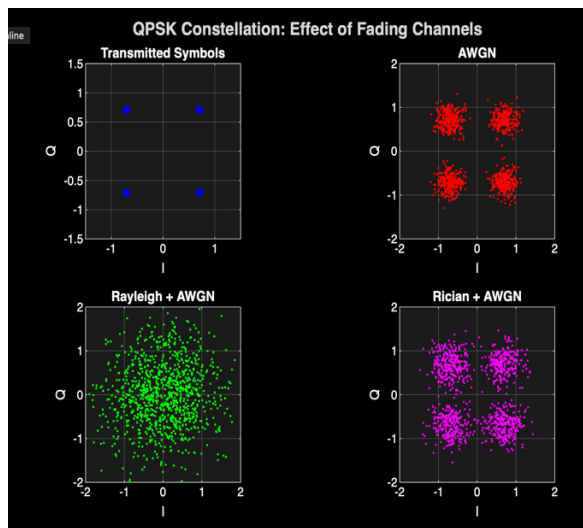
Under Rayleigh, the performance degrades significantly, while Rician ($K = 5$) improves results thanks to its LOS component.

5.8 BPSK Constellation under Fading



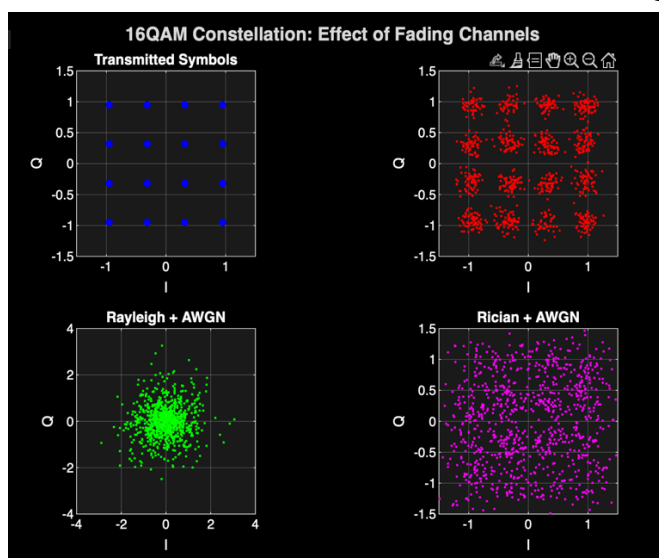
BPSK is the most noise-resistant modulation scheme. Even in fading environments, the receiver can clearly distinguish between the two states, which explains its excellent BER performance in earlier graphs.

5.9 QPSK Constellation under Fading



QPSK provides a good balance between reliability and data rate. It is more efficient than BPSK but still robust enough to perform well under fading, especially when a line-of-sight component (high K-factor) is present.

5.10 16-QAM Constellation under Fading



16-QAM achieves high data rates but is more easily distorted by fading compared to simpler schemes like BPSK or QPSK. It performs best when the channel is strong and stable such as in Rician or AWGN conditions.

6. Conclusion and Future Work

6.1 Conclusion

This project successfully simulated an adaptive modulation system in MATLAB Online using BPSK, QPSK, and 16-QAM over AWGN, Rayleigh, and Rician channels.

The Monte-Carlo results showed that the system can dynamically adjust modulation based on SNR, maintaining a target BER of 10^{-3} while maximizing spectral efficiency.

At low SNR, BPSK ensured reliable transmission; at medium SNR, QPSK provided balanced performance; and at high SNR, 16-QAM achieved the highest data rate.

Overall, the adaptive logic worked effectively, proving that such modulation is ideal for 5G, SDR, and IoT systems that require real-time adaptation to changing channel conditions.

6.2 Future Work

While the simulation successfully demonstrates adaptive modulation principles, several improvements can be made in future studies:

1. **Higher-Order Modulations** - Extend the model to include 64-QAM and 256-QAM to analyze performance in ultra-high data-rate environments.
2. **Error-Correction Coding** - Integrate Forward Error Correction (FEC) techniques such as convolutional or LDPC codes to further reduce BER under severe fading.
3. **Mobility and Doppler Effects** - Simulate user mobility and time-varying channels to study the impact of Doppler spread on adaptive decisions.
4. **Hardware Implementation** - Deploy the algorithm on Software-Defined Radio (SDR) platforms (e.g., USRP or GNU Radio) for real-time testing.
5. **Machine-Learning-Based Adaptation** - Replace fixed SNR thresholds with AI-driven adaptive decision models that can learn optimal switching patterns from live channel data.

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